



Syllabus Application



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CS 300 Data Structures

Faculty Faculty of Engineering and Natural Sciences

Semester Spring 2025-2026

Course CS 300 - Data Structures

Time/Place	<u>Time</u>	<u>Week Day</u>	<u>Place</u>	<u>Date</u>
	16:40-17:30	Mon	FASS-G062	Feb 16-May 22, 2026
	12:40-14:30	Tue	UC-G030	Feb 16-May 22, 2026

Level of course Undergraduate

Course Credits SU Credit:3, ECTS:6, Basic:1, Engineering:5

Prerequisites CS 204

Corequisites CS 300R

Course Type Lecture

Program Requirements >

Instructor(s) Information

Saima Gül

Email: saima.gul@sabanciuniv.edu

Course TAs and LAs Office Hours:

[OFFICE HOURS](#)

Course Information

Catalog Course Description

This course covers some of the fundamental data structures as well as the algorithms operating on them. Topics included are: Introduction to theoretical aspects of computing: modeling algorithms and their run times, computational complexity. Linear data structures (lists, stacks, queues), trees (binary search trees, AVL trees, tree traversals), hash tables, priority queues, disjoint sets, sorting algorithms, graphs and their representations, graph algorithms (depth first and breadth first search, single source shortest path algorithms), sorting algorithms, algorithmic paradigms (divide and conquer, greedy, dynamic programming).

Course Learning Outcomes:

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|----|--|
| 1. | describe common data structures and associated algorithms |
| 2. | contrast the advantages and disadvantages of these data structures |
| 3. | apply these data structures to develop algorithmic solutions to practical problems |
| 4. | design new data structures and associated algorithms |
| 5. | analyze the computational complexity of simple algorithms and programs |

Course Objective

To introduce fundamental techniques of algorithm analysis

To introduce common data structures, their properties and implementation

To apply C++ programming skills to implement solutions using common data structures to practical problems

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Course Materials

Resources:

Data Structures & Algorithm Analysis in C++, (2nd, 3rd, or 4th edition), Mark Allen Weiss, Pearson, 2014, ISBN 0-273-76938-3 (4th edition).

Technology Requirements:

Assessment

The final grade for this course will be based on:

	Percentage (%)	Number of assessment methods
Final	35%	
Midterm	45%	2
Homework	20%	5

Exam Schedule:

<u>Time</u>	<u>Event</u>	<u>Details</u>
Mar 31, 2026 12:40	Midterm: MidTerm-1	
May 9, 2026 13:00	Midterm: MidTerm-2	

Policies

Course Policies:

Attendance:

Throughout the course, the instructor may cover material beyond the textbook and Lecture slides. Accordingly, exam questions may be based on any topic discussed during lectures or recitation sessions. Students are therefore strongly advised to attend all classes regularly.

Make-Up:

1. No makeups for missed homeworks.

2. If you miss one exam (either a midterm or the final) due to a documented valid excuse (e.g., an official medical report), and the instructor is notified within 48 hours of the exam, you will be permitted to take a make-up exam after the final examination period.

Students may not miss more than one exam. If more than one exam is missed, the make-up exam will replace only the highest-weighted missed exam, and all remaining missed exam(s) will receive a grade of zero.

3. Make-up exams are often more comprehensive and cover all topics taught in the course from start to end.

Late Submission:

Late submission will be accepted for up to one day, with a penalty of 10 points.

Academic Integrity:

Sharing work or submitting similar solutions (with other students or AI) is considered plagiarism. Plagiarism cases will be referred to the Dean's Office for disciplinary action.

Special Notes:

Students are responsible for announcements made in lectures, SUCourse+, and email.

Regularly check your Sabancı University email.

Failure to follow announcements is not a valid excuse.

Use of AI:

It is strongly recommend that you avoid using AI tools (such as ChatGPT) as much as possible and use them very responsibly.

You may use AI tools to enhance your learning and improve your understanding of the course material; however, generating complete homework solutions using these tools is not permitted. Homework submissions found to rely excessively on LLM-generated content may receive a grade of zero and/or require the student to complete an oral examination.

The use of LLMs or any AI tools during quizzes or exams, or to generate complete homework submissions, is strictly prohibited. Such use constitutes a violation of academic integrity and will result in disciplinary action if detected.

AI Policies

[Guidelines on the "Use of Generative AI" for students](#)

SU Academic Integrity Statement:

<https://www.sabanciuniv.edu/en/academic-integrity-statement>

Health Report Requirements

[Student Medical Reports Instruction Letter](#) 

Weekly Content

Week	Topic
Week 1	Overview
Week 2	Algorithm Analysis
Week 3	Linear Data Structures 1 (Linked List)
Week 4	Linear Data Structures 2 (Stacks, Queues)

Week	Topic
Week 5	Trees 1
Week 6	Trees 2
Week 7	Hash Tables
Week 8	Heaps (Priority Queues)
Week 9	Sorting 1
Week 10	Sorting 2
Week 11	Disjoint Sets
Week 12	Graph Data Structures 1
Week 13	Graph Data Structures 2
Week 14	Algorithm Design Techniques